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ODD END SEMESTER EXAMINATION DECEMBER-2024

Name of the Course: B.Tech

Semester: III

Name of the Course: Microprocessors & Microcontroller Course Code: TME 311

Time: 3Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1	(10marks)	
(a)	Perform the following operation (i) $(736)_8 + (565)_8 = ()_8$ (ii) $(9ABC)_2 - (DAD)_{16} = ()_{16}$ (iii) $(1011)_2 + (1000)_2 = ()_2$ (iv) $(1011)_2 - (1000)_2 = ()_2$	CO2
(b)	Convert the following: (i) $(23.23)_8 = ()_2 = ()_{10} = ()_{16}$ (ii) $(456)_{10} = ()_2 = ()_8 = ()_{16}$ (iii) $(1011.01)_2 = ()_8 = ()_{10} = ()_{16}$	
(c)	Explain (i) BCD code (ii) Gray code	
Q2	(10marks)	
(a)	Write a short note on evolution of microprocessor from Intel 8085 to Intel pro Pentium series.	CO2
(b)	Draw and explain the internal architecture of 8085 microprocessor.	
(c)	Explain the function of various pins available in 8085 microprocessor: (i) RST 7.5 (ii) HLDA (iii) READY (iv) ALE (v) INTR	
Q3	(10marks)	
(a)	Write down the differences between I/O mapped and memory mapped I/O interfacing.	CO1, CO2
(b)	Write a short note on memory interfacing. Explain partial and absolute decoding.	
(c)	Explain the classification of registers available in 8085 microprocessor.	

Q4	(10marks)	
(a)	Write differences between I/O mapped and memory mapped I/O interfacing.	CO2
(b)	Enlist the following instructions according to 8085 addressing modes (i) LXI (ii) RLC (iii) XOR (iv) LDAX (v) SBB	
(c)	Explain the following instructions with suitable examples (i) XTHL (ii) LHLD (iii) SHLD (iv) SPHL (v) STC	
Q5	(10 Marks)	
(a)	How many types of registers are there in 8085 microprocessor? Explain with the help of examples.	CO1, CO2
(b)	OR	
(c)	Perform the subtraction using 2's compliment method. (i) $(10110010)_2 - (10001011)_2 = ()_2$ (ii) $(10001011)_2 - (10110010)_2 = ()_2$	